Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

Online Appendix

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A Reduced-form stress tests and prior elicitation

This appendix describes how we set the priors on the climate shifters $(\rho_s^{SST}, \rho_s^{CHL})$ that enter the state-space model of the Stock dynamics subsection of the main text, together with the identifiability diagnostic that motivates the use of Bayesian methods over direct maximum likelihood.

A.1 Deterministic hindcast and cross-variant fit

For each stock s we estimate a deterministic Schaefer hindcast over 2000–2024, $\hat{B}_{s,t+1} = \hat{B}_{s,t} + r_{s,t}\hat{B}_{s,t}(1-\hat{B}_{s,t}/K_s)-C_{s,t}$ with $r_{s,t} = r_s^0 \exp(\rho_s^{SST}(SST_{t-1}-\overline{SST})+\rho_s^{CHL}(\log CHL_{t-1}-\overline{\log CHL}))$, stock-dynamics parameters $(r_s^0, K_s, B_{0,s})$ held fixed at the IFOP/SPRFMO single-species prior means reported in Table A.1, and shifters estimated by bounded least squares with each shifter constrained to $[-3, 3]$,[1] minimising median absolute percent error (MAPE). The structural priors

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[1]The bound is a biologically extreme upper limit: $|\rho| = 3$ implies that a one-unit anomaly changes productivity by roughly 95%.

draw on the FishLife meta-analysis for intrinsic growth rates (Thorson 2020), the IFOP V–X stock-assessment informes for anchovy and sardine carrying capacity (Instituto de Fomento Pesquero 2024, 2022), and the range-wide SPRFMO assessment for jack mackerel (South Pacific Regional Fisheries Management Organisation 2025).

Table A.1: Single-species priors on structural stock-dynamics parameters used by the deterministic hindcast and as priors in the state-space likelihood. Priors are log-normal on $(r_s^0, K_s, B_{0,s})$; the entry $\log \mathcal{N}(\mu, \sigma)$ denotes $\log \theta \sim \mathcal{N}(\log \mu, \sigma)$.

Stock	r_s^0 (yr^{-1})	K_s (kt)	$B_{0,s}$ (kt)
Anchovy	$\log \mathcal{N}(0.6,\ 0.25)$	$\log \mathcal{N}(2{,}200,\ 0.15)$	$\log \mathcal{N}(376,\ 0.10)$
Sardine	$\log \mathcal{N}(0.9,\ 0.25)$	$\log \mathcal{N}(3{,}000,\ 0.15)$	$\log \mathcal{N}(496,\ 0.10)$
Jack mackerel	$\log \mathcal{N}(0.35,\ 0.25)$	$\log \mathcal{N}(8{,}000,\ 0.25)$	$\log \mathcal{N}(5{,}889,\ 0.15)$

Notes. Intrinsic growth-rate priors (r_s^0) follow Thorson's FishLife meta-analysis. Carrying capacity (K_s) for anchovy and sardine is derived from the unfished spawning biomass B_{D0} reported in the IFOP V–X stock-assessment informes (anchovy 2024, sardine 2022), scaled to total biomass; for jack mackerel, K_s is re-scaled from the range-wide SPRFMO 2025 assessment. Initial-biomass priors ($B_{0,s}$) are not virgin biomass: they represent the state-space initial condition at $t = 2000$ and are centred at the first observed value in the window (total biomass for the coastal stocks, IFOP acoustic survey for jack mackerel). This empirical-Bayes choice is necessary for identification of the initial state under $N = 25$ annual observations; see Online Appendix A.2.

Four nested variants are reported: base ($\rho = 0$), +SST only, + logCHL only, and joint +SST+ logCHL.

Table A.2: Median absolute percent error of the deterministic Schaefer hindcast across shifter variants.

Stock	Base	+ SST	+ log CHL	+ SST and log CHL
Anchovy	95.9	30.8	95.1	26.6
Sardine	98.5	23.1	98.5	20.5
Jack mackerel	44.2	44.3	49.1	48.5

No variant brings any of the three stocks below a 20% MAPE threshold. The two coastal stocks benefit from activating SST (anchovy: 95.9% → 30.8%, sardine: 98.5% → 23.1%), with the joint specification delivering further marginal improvement. Jack mackerel hindcast errors remain in the 44–49% range across all variants, with no shifter distinguishably better than the base.

The point estimates from the joint variant are interpretable for the coastal stocks. For anchovy both shifters are negative ($\rho^{SST} \approx -2.3$, $\rho^{CHL} \approx -2.3$); for sardine $\rho^{SST} \approx -2.0$ is negative but $\rho^{CHL} \approx +2.1$ is positive—the opposite sign on CHL across the two species reflects underlying ecological differences. For jack mackerel $\rho^{SST} \approx +0.6$ is small and ρ^{CHL} binds at the lower boundary; we read this as a failure of identification at the Centro-Sur scale rather than as strong climate forcing.

A.2 Identifiability diagnostic and prior elicitation

Two diagnostics motivate the Bayesian approach. First, the MLE's difficulties are not due to collinearity: SST and CHL are uncorrelated over the 2000–2024 sample (Pearson $r = 0.030$, $p = 0.888$), so the difference in hindcast error across variants cannot be blamed on linear dependence between the shifters. Second, the jack mackerel MLE binds at the boundary: the deterministic hindcast cannot separate the climate signal from noise, so ρ is not separately identifiable. Together these point to a fundamental identification problem under $N = 25$ observations, which the Bayesian state-space specification addresses in a single estimation step.

Table A.3 reports the priors on $(\rho_s^{SST}, \rho_s^{CHL})$ used in the structural likelihood. Prior centres for the two coastal stocks are the joint MLEs rounded to one decimal place; jack mackerel priors are centred at zero so that posterior identification is driven by the structural likelihood rather than by the boundary solution. All priors are independent normal with unit standard deviation. We do not impose hierarchical pooling across species because ρ^{CHL} has opposite signs for anchovy and sardine—pooling would shrink the two estimates toward a common center and erase this ecological distinction.

This prior-elicitation strategy is a form of empirical Bayes: the prior centres for anchovy and sardine come from the same 2000–2024 biomass series that enters the likelihood. Standard Bayesian practice keeps prior and data independent; we accept this departure because with $N = 25$ annual observations it is necessary for identification.

We adopt this trade-off deliberately for two reasons. First, the prior $\sigma = 1$ is wide enough that the data move the posterior substantially when informative: for anchovy, the posterior of ρ^{CHL} migrates to -3.64, more than one prior standard deviation away from the prior centre of -2.3 (see the posterior-identification table of the main text for full posterior summaries). Second, centring all priors at zero under $N = 25$ would leave the joint posterior of $(r_s^0, K_s, B_{0,s}, \rho_s^{SST}, \rho_s^{CHL})$ poorly identified, producing flat posteriors regardless of whether climate coupling is real or absent. The empirical-Bayes choice therefore makes the non-identification result for jack mackerel a substantive finding, not a numerical artefact.

Table A.3: Independent normal priors on the climate shifters used in the state-space likelihood.

Stock	$\mu_{\rho^{SST}}$	$\sigma_{\rho^{SST}}$	$\mu_{\rho^{CHL}}$	$\sigma_{\rho^{CHL}}$	Source
Anchovy	-2.3	1.0	-2.3	1.0	Joint MLE
Sardine	-2.0	1.0	+2.1	1.0	Joint MLE
Jack mackerel	0.0	1.0	0.0	1.0	Boundary solution; vague prior

B Predictive diagnostics

This appendix reports the nested predictive comparison referenced in the Stock dynamics subsection of the main text, comparing three state-space specifications: (i) species-independent dynamics without climate shifters (*ind*), (ii) the same with a cross-stock residual covariance Ω (*omega*), and (iii) the full model augmented with the climate shifter equation of the Stock dynamics subsection of the main text (*full*). Two predictive metrics are reported. PSIS-LOO (Pareto-smoothed importance-sampling leave-one-out) is a Bayesian analogue of cross-validation that estimates how well each model would predict each observation if that observation were held out of the sample. LFO (leave-future-out) is its time-series counterpart: the model is refit on observations up to year t and evaluated on the one-step-ahead predictive density at $t + 1$. Both metrics report Expected Log Predictive Density (ELPD); higher values indicate better predictive performance.

PSIS-LOO favours the climate-augmented specification with a positive ΔELPD of 14–18 relative to the climate-free specifications, indicating that the in-sample predictive performance improves when the climate shifter is included.

The one-step-ahead LFO comparison is essentially a tie across the three specifications, indicating

Table B.1: PSIS-LOO comparison across state-space specifications.

Model	$\widehat{\text{ELPD}}_{\text{loo}}$	SE	$\Delta\widehat{\text{ELPD}}$	SE_{Δ}	$\hat{p}_{\text{loo}}$
full	-8.41	8.28	–	–	64.8
ind	-23.00	8.01	-14.59	4.93	62.4
omega	-26.43	9.16	-18.02	5.59	57.9

Note:
ΔELPD is the difference relative to full (the leading model); SE is the standard error of that difference.

Table B.2: One-step-ahead LFO comparison across state-space specifications.

Model	$\widehat{\text{ELPD}}_{\text{LFO}}$	Mean	$\Delta\widehat{\text{ELPD}}$
full	-29.10	-2.08	–
ind	-29.46	-2.10	-0.37
omega	-30.46	-2.18	-1.36

Note:
ELPD sums one-step-ahead predictive log densities over 14 target years per model, covering five origin years equally spaced from 2011 to 2022 (roughly half of the 2000–2024 window) across three stocks. ΔELPD is the difference relative to full (the leading model).

that the climate-augmented specification does not improve one-step-ahead forecasts relative to the climate-free alternatives. This reflects sample size, not a flaw in the climate specification: with only 25 years of biomass data, the climate shifters capture how SST and CHL modulate stock productivity, and their effect accumulates over multiple years; in any single year it is small relative to process and observation noise, so the climate signal is identifiable across the full 25-year window but not in one-step-ahead increments. The climate-augmented specification remains the primary model reported in the main text.

C Posterior-predictive checks

This appendix reports posterior-predictive diagnostics of the full state-space specification: (i) the smoothed latent biomass versus the observed survey-based estimates, and (ii) the corresponding year-level residuals.

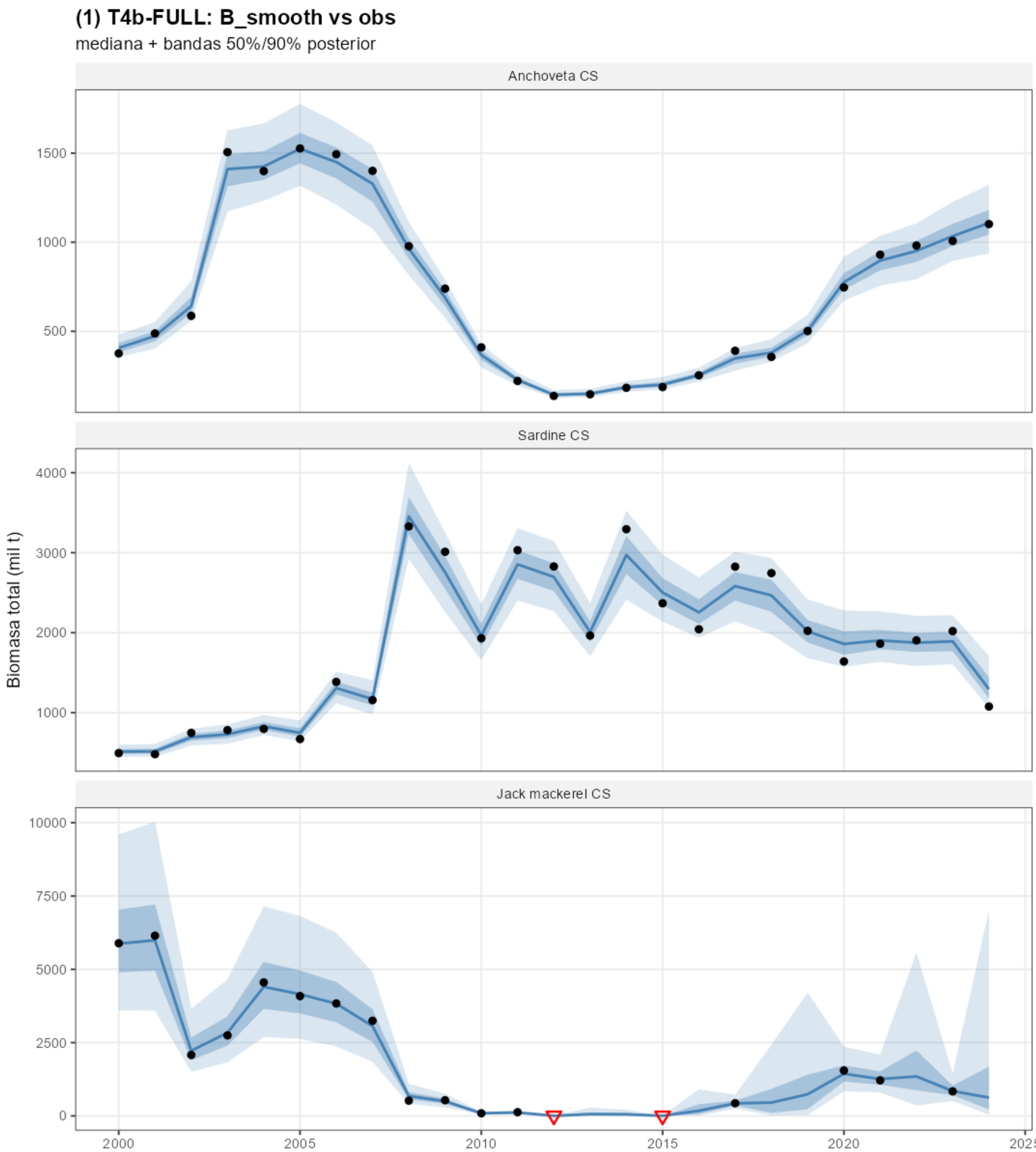

Figure C.1: Smoothed latent biomass $B_{i,t}$ of the full specification against survey-based SSB observations, by stock. Solid line: posterior median; shaded band: 90% credible interval. Open circles: official IFOP/SPRFMO point estimates; downward arrows: left-censored observations at the assessment detection limit (jack mackerel 2012, 2015).

The smoothed band has a median width of approximately 20% of the posterior mean, tightening over years with direct observation and widening across the seven non-surveyed jack mackerel years (latent biomass identified dynamically through the Schaefer transition and two censored observations in 2012 and 2015).

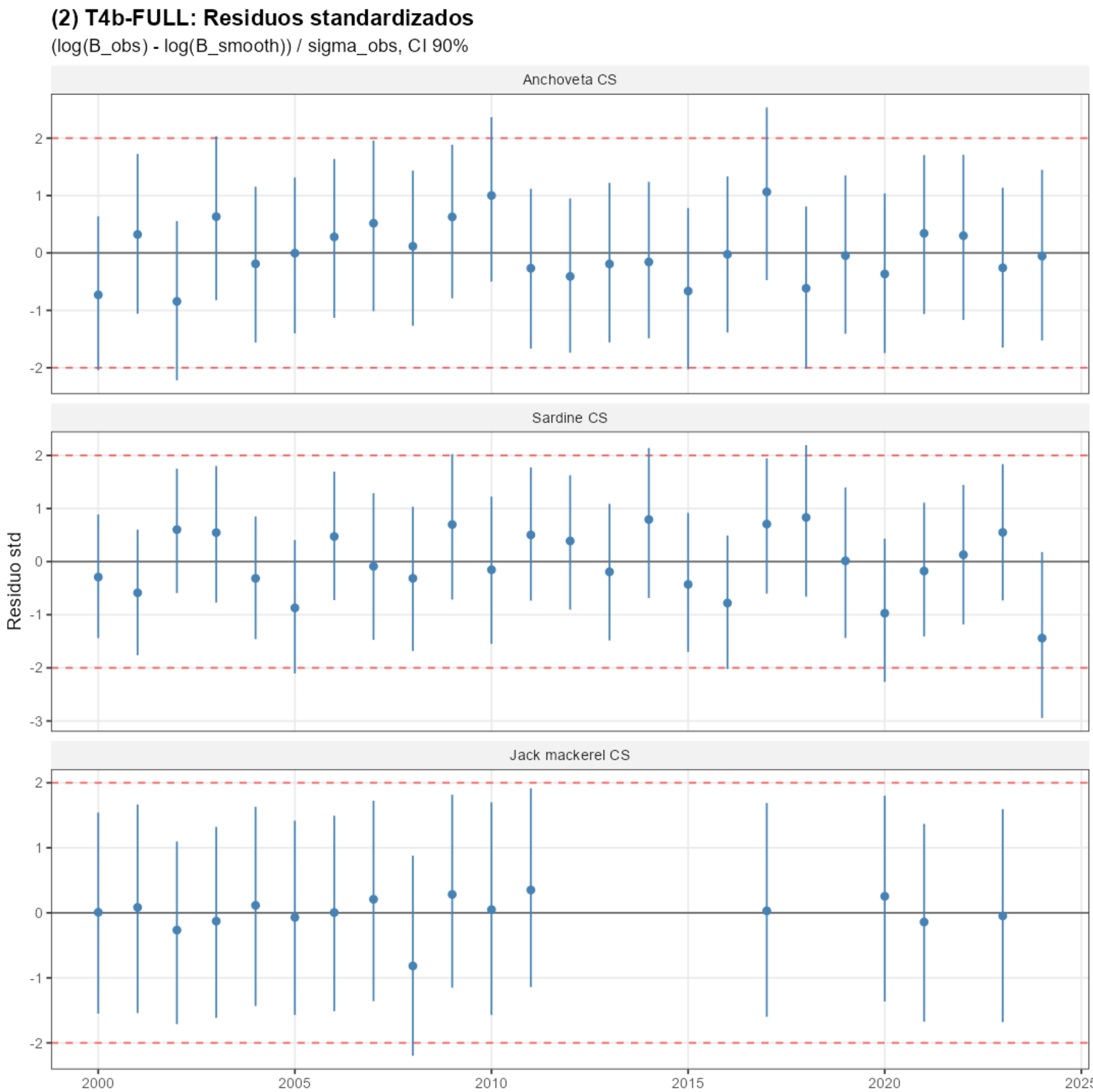

Figure C.2: Standardised year-level residuals of the full specification by stock. The Ljung–Box test of residual autocorrelation does not reject the null at conventional levels for any of the three stocks.

D Markov-chain convergence diagnostics

Table D.1 reports the Gelman–Rubin $\hat{R}$ and the bulk/tail effective sample sizes for the top-level parameters of the full state-space specification, evaluated on four independent Hamiltonian Monte Carlo chains of 2,000 post-warmup iterations each. All $\hat{R}$ values are below 1.01 and all effective sample sizes exceed 400, indicating that posterior summaries are not contaminated by between-chain divergence or by autocorrelation between adjacent draws.

Table D.1: Markov-chain convergence diagnostics for the top-level parameters of the T4b-full state-space specification.

Parameter	N	Max $\widehat{R}$	Min ESS_{bulk}	Min ESS_{tail}
r_i^0	3	1.001	8,446	10,508
K_i	3	1.000	9,789	9,777
$B_{i,0}$	3	1.001	18,919	11,844
$\sigma_{\text{proc},i}$	3	1.003	3,353	5,997
$\sigma_{\text{obs},i}$	3	1.009	1,370	936
ρ_i^{SST}	3	1.000	8,312	10,147
ρ_i^{CHL}	3	1.001	14,319	10,633
Ω (off-diagonal)	3	1.001	5,732	8,544
All top-level parameters	24	1.009	1,370	936

Note:
Diagnostics are computed from four chains of 2,000 post-warmup iterations. Within each parameter family, the table reports the worst case across the three stocks (or the three off-diagonal entries of Omega): the largest split-Rhat and the smallest bulk- and tail-effective sample sizes. The diagonal of Omega is fixed at one and is therefore excluded. The conventional thresholds are Rhat <= 1.01 and ESS >= 400 for all top-level parameters.

E Spatial robustness of the jack mackerel non-identification result

The non-identification of $(\rho^{SST}_{\text{jack mackerel}}, \rho^{CHL}_{\text{jack mackerel}})$ reported in the Identification of climate shifters section of the main text admits two readings: a true null (jack mackerel productivity has no climate response at the annual aggregate) or a measurement gap (the relevant forcing operates at a scale broader than the local shifters capture). This appendix first establishes that the sample places jack mackerel at the edge of detection for a climate shifter of magnitude comparable to the identified coastal-stock shifters, then tests five mechanical explanations for the null—spatial aggregation, joint specification, biomass coverage, basin-scale forcing, and SPRFMO integration—and concludes that the null is structural within the 2000–2024 Centro-Sur record.

E.1 Identification power on the sample

Before testing alternative specifications, we check whether the sample is informative enough to identify a jack mackerel climate shifter of magnitude comparable to the coastal stocks. The minimum shifter detectable at 80% power for a two-sided 90% credible interval is $|\rho_i^X|_{\min} \approx 2.56\,\mathrm{SE}(\hat{\rho}_i^X)$, where $\mathrm{SE}(\hat{\rho}_i^X) \approx \sigma_{i,\mathrm{resid}}/[\mathrm{sd}(X)\sqrt{N-K}]$ from a linearisation of the Schaefer transition with $\sigma_{i,\mathrm{resid}} = \sqrt{\sigma_{i,\mathrm{proc}}^2 + 2\sigma_{i,\mathrm{obs}}^2}$ the residual noise (combining process and observation noise), $N = 24$ lag-one observations, and $K = 3$ regressors.

At posterior medians, the residual noise is $\sigma_{\mathrm{resid}} = 0.28$ for anchovy, 0.25 for sardine, and 1.28 for jack mackerel—roughly 4.6 times higher for jack mackerel than for the coastal stocks. This is the main reason the prior dominates the jack mackerel posterior in the full state-space fit. Table E.1 reports the detectable $|\rho|_{\min}$ for each stock and candidate shifter. For the coastal stocks, $|\rho|_{\min}$ ranges from 0.26 to 0.33 across all shifters, well below the identified posterior magnitudes reported in the main text ($|\rho|$ between 1.06 and 3.64), confirming that the coastal-stock identification is supported by ample power. For jack mackerel, $|\rho|_{\min}$ ranges from 0.71 (basin-scale ENSO) to 1.50 (local SST)—of the same order of magnitude as the coastal-stock estimates. A jack mackerel climate response of coastal-stock magnitude would therefore be at the edge of detection on the present sample; a smaller response would not be detectable at all. The direct test of whether such a response is present is the posterior-to-prior comparison in the next subsection, to which we now turn.

Table E.1: Minimum shifter magnitude $|\rho|_{\min}$ detectable at 80% power on the 2000–2024 sample. $\sigma_{\text{resid}} = \sqrt{\sigma^2_{\text{proc}} + 2\sigma^2_{\text{obs}}}$ evaluated at posterior medians.

Stock	σ_{resid}	$\lvert\rho_{SST}\rvert_{\min}$	$\lvert\rho_{\log\text{CHL}}\rvert_{\min}$	$\lvert\rho_{\text{ENSO}}\rvert_{\min}$
Anchovy	0.28	0.33	0.29	—
Sardine	0.25	0.29	0.26	—
Jack mackerel	1.28	1.50	1.32	0.71

Notes. Detectable magnitudes computed from $|\rho_i^X|_{\min} \approx 2.56\,\sigma_{i,\text{resid}}/[\text{sd}(X)\sqrt{N-K}]$ with $N = 24$, $K = 3$. sd(X) are sample standard deviations of the Centro-Sur EEZ-aggregated SST and log CHL anomalies and of the Niño 3.4 index over the estimation window.

E.2 Alternative specifications

We re-fit the state-space jack mackerel block under five alternative shifter configurations, each designed to test one mechanical explanation for the local null. Ratios of posterior-to-prior standard deviation at unity indicate no posterior updating relative to the prior; all configurations are summarised in Table E.2 below.

Spatial extent. The local SST and log~CHL shifters are re-computed by aggregating environmental anomalies over three progressively wider domains: $\mathcal{D}_1$ (Chilean EEZ, 18–42°S out to 200 nautical miles), $\mathcal{D}_2$ (extended offshore to 1,000 km) and $\mathcal{D}_3$ (Southeast Pacific, 5–45°S and 70–120°W). Spatial averages are weighted by grid-cell area and expressed as anomalies from the 2000–2024 mean.

Biomass coverage (dual-source extension). The Centro-Sur and Northern Chilean acoustic biomass series exhibit a 0.88 log-scale correlation across overlapping years (2012–2024), motivating a joint-likelihood specification in which a common shifter ρ^{common} is identified simultaneously on both records. The extension multiplies the number of informative jack mackerel-biomass observations by approximately 1.7 while preserving the prior structure of the main fit.

Basin-scale forcing (ENSO). Replacing the local coastal shifters with the basin-scale ENSO Niño 3.4 index tests whether the relevant forcing operates at the basin scale, consistent with Peña-Torres et al. (2017) (ENSO-driven shifts in jack mackerel location choices) and Arcos et al. (2001)

(SPRFMO-scale stock dynamics under broad-basin forcing). The specification is run at lag~1 and lag~2.

SPRFMO integration. Combining the Centro-Sur record with the SPRFMO range-wide assessment (Chile, Peru, Ecuador, and adjacent high-seas) fails to converge, so the two cannot be fit under a single shifter.

Joint specification. All three shifters (local SST, local log~CHL, basin-scale ENSO at lag~1) are estimated simultaneously for jack mackerel, holding anchovy and sardine at the main specification.

Table E.2: Posterior-to-prior standard deviation ratios of the jack mackerel climate shifters under five alternative specifications.

Test	Specification	σ(SST)	σ(log CHL)	σ(ENSO)
Spatial extent	EEZ $\mathcal{D}_1$	0.998	1.014	—
	Offshore-extended $\mathcal{D}_2$	1.002	1.009	—
	Southeast Pacific $\mathcal{D}_3$	0.999	1.011	—
Biomass coverage	Dual-source (Centro-Sur + Northern acoustic)	0.94	0.99	—
Spatial scale	Basin-scale ENSO Niño 3.4, lag 1	—	—	0.98
	Basin-scale ENSO Niño 3.4, lag 2	—	—	1.01
Joint specification	Local SST + local log CHL + ENSO (lag 1)	1.03	1.00	0.98

Notes. All entries are $\sigma_{\text{post}}/\sigma_{\text{prior}}$. Anchovy and sardine ratios remain stable at 0.43–0.83 across all configurations.

All five tests deliver a posterior-to-prior ratio between 0.94 and 1.03 for the jack mackerel shifters. The ratio measures how much the data shrink the prior: a value near 1 means the posterior is no narrower than the prior (no information), and a value below 1 means the data refine the prior. For comparison, the anchovy and sardine shifters identified in the main text have ratios between 0.43 and 0.83, with posterior medians moving substantially away from their prior centres. For jack mackerel, the ratios near unity imply that the posterior medians stay near the prior mean (zero) with credible intervals nearly spanning the prior range, under every specification. This is exactly what the power calculation of Table E.1 predicts: a jack mackerel climate shifter smaller than the detection threshold $|\rho|_{\min} = 0.71$–1.50 would leave no detectable signal in the posterior. Propagating the posterior of $\rho^{\text{ENSO}}_{\text{jack mackerel}}$ through the CMIP6 ensemble yields a 90% predictive interval for the jack mackerel productivity factor spanning roughly three orders of magnitude ($[0.05, 19.5]$) at SSP5-8.5

end-of-century, validating the fixed-$r^{\star}_{\text{jack mackerel}}$ convention of the main text.

E.3 Implication for the main claim

The five tests above rule out the natural mechanical explanations for the jack mackerel non-identification: the result does not respond to broader spatial aggregation of the local shifters ($\mathcal{D}_1$ through $\mathcal{D}_3$), to joint specification, to expanded biomass coverage (dual-source), to basin-scale forcing (Niño 3.4), or to integration with the SPRFMO range-wide assessment. The power calculation (Table E.1) places the jack mackerel detection threshold at the edge of the coastal-stock shifter magnitudes: a coastal-magnitude effect would be barely detectable on the present sample, and a smaller effect not at all. The convergence of these findings points to a structural limitation of the present record. Closing the gap will require either a substantially longer biomass record or a basin-scale assessment with explicit cross-stock spillover. Within the present record, holding $r^{\star}_{\text{jack mackerel}}$ fixed at the prior mean is the defensible reporting choice and is the convention adopted throughout the main text.

F Inter-sectoral cessions

This appendix documents the realised inter-sectoral quota cession flow over 2013–2024, the single administrative re-allocation channel admitted by the de jure fractioning regime. We extract assigned, ceded, effective, and captured tonnage by sector for each species–year from the annual SERNAPESCA quota control reports.

Two qualitative facts emerge from the consolidated 2013–2024 series, separated by species pattern: anchovy and sardine in Table F.1, jack mackerel in Table F.2. First, in the anchovy and sardine fisheries the industrial sector cedes a large share of its assigned TAC to artisanal cessionaries in every year of the window: the share transferred ranges from 12% to 72% over 2013–2017 and tightens to 85%–$\sim$100% over 2018–2024, scaling with the TAC from 26 to 50 kt per year in anchovy and 51 to 77 kt per year in sardine. Second, in the jack mackerel fishery the cession flow is an order of magnitude smaller relative to the assigned TAC—never exceeding 14% in any year and falling below 5% in eight of the twelve years—and reverses direction across years.

The post-cession allocation reflects the joint pre-reform fractioning and cession regime rather than the de jure fractioning alone. Whether industrial quota holders would continue to cede 85–99% of their anchovy and sardine TACs under a collapsing biomass is an open question. The contracts depend on the productivity of the ceded stocks, and the cession volumes observed over 2013–2024 reflect TAC and price conditions that the projection horizon will not reproduce.

Table F.1: Inter-sectoral cession flow, anchovy and sardine, Valparaíso to Los Lagos, 2013–2024. Source: SERNAPESCA quota control reports.

Year	Species	IND assigned (kt)	IND cession (kt)	Share ceded
2013	anchovy	25.5	−3.1	12%
2013	sardine	128.5	−31.1	24%
2014	anchovy	7.7	−3.9	51%
2014	sardine	104.8	−46.6	44%
2015	anchovy	6.3	−2.5	40%
2015	sardine	65.2	−21.8	33%
2016	anchovy	7.3	−2.1	29%
2016	sardine	59.8	+15.6	26%
2017	anchovy	12.6	−9.0	72%
2017	sardine	72.4	−48.9	67%
2018	anchovy	12.8	−10.7	84%
2018	sardine	74.2	−63.2	85%
2019	anchovy	27.4	−25.8	94%
2019	sardine	76.0	−69.6	92%
2020	anchovy	38.6	−33.3	86%
2020	sardine	67.2	−60.9	91%
2021	anchovy	45.3	−45.1	∼100%
2021	sardine	76.9	−76.5	∼100%
2022	anchovy	52.2	−50.5	97%
2022	sardine	77.7	−71.6	92%
2023	anchovy	39.9	−37.1	93%
2023	sardine	63.3	−61.7	98%
2024	anchovy	53.8	−45.5	85%
2024	sardine	53.9	−51.0	95%

Notes. Industrial share of Cuota Global de Captura, Valparaíso to Los Lagos. Negative cession denotes industrial-to-artisanal flows.

Table F.2: Inter-sectoral cession flow, jack mackerel, 2013–2024. Source: SERNAPESCA quota control reports.

Year	Species	IND assigned (kt)	IND cession (kt)	Share ceded
2013	jack mackerel	138.4	+0.5	0%
2014	jack mackerel	198.6	+27.3	14%
2015	jack mackerel	203.1	−17.2	8%
2016	jack mackerel	203.0	+11.7	6%
2017	jack mackerel	228.3	+9.7	4%
2018†	jack mackerel	326.9	+16.1	5%
2019†	jack mackerel	339.0	+12.0	3%
2020†	jack mackerel	1268.1	+11.8	1%
2021†	jack mackerel	448.7	+13.3	3%
2022†	jack mackerel	516.4	+15.2	3%
2023†	jack mackerel	637.0	−0.7	0%
2024†	jack mackerel	728.6	−31.5	4%

Notes. Conventions as in Table F.1. †2018–2024 is the national aggregate (the SERNAPESCA summary sheet is not disaggregated by sub-macrozone); the share-ceded column remains comparable because the additional Arica-to-Atacama volume enters both numerator and denominator.

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